# Paleomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: the Stark Formation

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Paleomagnetic results from 55 sampling sites throughout the Stark Formation are reported. The known stratigraphic sequence of these sites enables the behaviour of the geomagnetic field in these remote times (1750 m.y.) to be elucidated. Two polarity reversals are identified and these represent potentially useful correlative features in strata devoid of index fossils. One of these is investigated in detail and indicates that behaviour of the geomagnetic field during polarity reversals was essentially the same in the early Proterozoic as it has been over the last few million years. The pole position (145° W, 15° S, dp = 3.5, dm = 6.9) lies far to the west of that anticipated from earlier results, implying further complexity of the North American polar wander curve. Possible alternatives to this added complexity are discussed.

Nous soumettons les résultats paléomagnétiques de 55 endroits échantillonnés à travers la formation Stark. La séquence stratigraphique de ces endroits est connue et nous permet d'élucider le comportement du champ magnétique dans ces temps anciens (1750 Ma). Deux inversions de polarité ont été identifiées qui représentent des critères de corrélation possiblement utiles dans les strates dénuées de fossils index. Une de ces inversions est étudiée en détail et indique que le champ géomagnétique au cours d'inversions de polarité était essentiellement le même durant le Protérozoique inférieur que ce qu'il a été durant les derniers millions d'années. La position du pôle (145° W, 15° S, dp = 3.5, dm = 6.9) est loin à l'ouest de ce à quoi on s'attendait à partir d'anciens résultats qui ajoutaient une certaine complexité à la courbe polaire de l'Amérique du Nord. Nous discutons quelques alternatives possible à cette complexité supplémentaire.

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## Introduction

On the Canadian Shield there are a number of basins containing great thicknesses of relatively fresh sedimentary and volcanic rocks of Precambrian age. Since the stratigraphy in some of these basins is well known and since some of the formations consist primarily of red beds, these basins provide good access to portions of the Precambrian paleomagnetic record for North America.

The present study is part of a program of paleomagnetic sampling of Lower Proterozoic rocks preserved in the East Arm Fold Belt of Great Slave Lake, N.W.T. This foldbelt lies within the Churchill Province and is bounded on the north by the Slave Province of the Canadian Shield. The nearby granites of the former province are Hudsonian in age (1450–2010 m.y.); those of the latter are of Kenoran age (2300–2640 m.y.) (Hoffman 1969). The 12 000 stratigraphic metres of sedimentary and volcanic rocks comprising the Great Slave Supergroup have been mapped by Stockwell (1936) and more recently by Hoffman (1968, 1969), whose nomenclature is followed here. Unconformably overlying the Great Slave Supergroup are strata of the Et-then Group, whereas the final geological element consists of diabase dykes and sills correlated with the Mackenzie igneous events (~ 1200 m.y.).

The Great Slave Supergroup has been subdivided by Hoffman (1968) into four groups: in order of decreasing age these are the Sosan, Kahochella, Pethei, and Christie Bay Groups. The Stark Formation is the lowest member of the Christie Bay Group, and it consists of at least 600 m of finely bedded red mudstones. Frequent discontinuous pale green layers and blotches up to a few centimetres thick give outcrops a striped appearance. Ripple marks, cross bedding, and hopper-shaped halite crystal casts are common. Hoffman (1969, p. 453) suggests that the Stark Formation was "deposited in a shallow sublittoral marine environment of high aridity".

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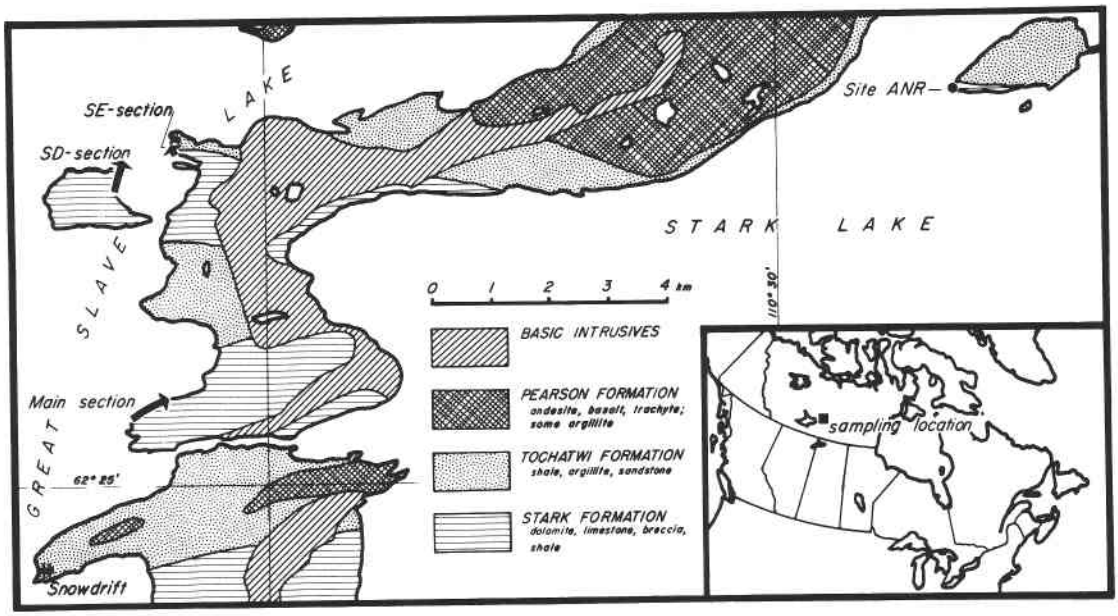



FIG. 1. Location map showing sampling locations. Heavy arrows point toward top of stratigraphic sections. Geology after Barnes (1951).

The age of Stark sedimentation can be inferred from its relationship to several radiometrically dated quartz diorite laccoliths, and also from the broader interpretation of the evolutionary history of the so-called Coronation geosyncline. For details, the reader should consult Hoffman (1968, 1969, 1973) and Hoffman *et al.* (1970); these authors conclude that the best available estimate of the age of the sedimentation phase to which the Stark Formation belongs is 1750 m.y. Conformably overlying the Stark Formation is the Tochatwi Formation, a paleomagnetic survey of which is reported in a companion paper (Evans and Bingham 1975).

## Methods

A total of 160 hand blocks were collected from 55 stratigraphically distinct sites. Forty-five of these represent a sequential sampling of a single 600 m thick section consisting of nearly vertically dipping beds along the shore of Christie Bay, Great Slave Lake (Fig. 1). The historical details of sampling this section have resulted in site names starting with SB, SC, SK, and AJ; this entire section is hereafter referred to as the "main section". The other stratigraphic sections are called the SD and SE sections, in accordance with the first two characters appearing in the site names.

At all but a few sites, beds that are predominantly red were sampled. Hand samples were collected because the generally friable nature of the rocks made field drilling impossible. For the most part, 3 blocks were collected at each site, but occasionally 2 or 4 blocks were taken; full details are given in Table 1. Orientation was by solar bearings and/or by geographical sight points tied to solar bearings.

The complete study involved about 1700 measurements of remanence which were made on a PAR model SM-1 spinner magnetometer. Routine partial demagnetization was carried out on all specimens used in the final analysis. Alternating field (AF) demagnetization was performed with the apparatus described by Murthy (1969), which is similar to that of McElhinny (1966). Thermal cleaning was accomplished using a small oven in a room-sized magnetic shield (Patton and Fitch 1962; Reid 1972); the magnetic field over the volume occupied by the specimens during thermal cleaning studies is less than 5 $\gamma$ and temperatures are controlled to $\pm 5$ °C.

## Results

### *The 'Main Section'*

Preliminary measurements showed that a reversal of magnetic polarity occurs approxi-

TABLE 1. Site mean paleomagnetic data for the Stark Formation

| Site | NRM | | | | | | T/H | CLEANED | | | | | | Strike | Dip |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *N* | *S* | *D* | *I* | *k* | *R* | | *N* | *S* | *D* | *I* | *k* | *R* | | |
| SEC | 3 | 3 | 181 | +15 | 18.6 | 2.8923 | 500 °C | 3 | 3 | **186** | **+17** | 163.2 | 2.9877 | 295 | 125 |
| | 3 | 3 | 183 | +15 | 64.7 | 2.9691 | 400 Oe | 3 | 3 | 187 | +21 | 156.7 | 2.9872 | | |
| SEB | 3[a] | 3 | * | * | 2.9 | 2.3200 | 500 °C | 3 | 4 | **190** | **+15** | 193.5 | 2.9897 | 292 | 126 |
| | 2[c] | 2 | 205 | −35 | 9.5 | 1.8946 | 800 Oe | 2 | 2 | 205 | +19 | 20.4 | 1.9510 | | |
| SEA | 3 | 3 | * | * | 1.9 | 1.9333 | 500 °C | 3 | 3 | * | * | 3.0 | 2.3336 | 285 | 128 |
| | 3 | 3 | * | * | 3.0 | 2.3364 | 800 Oe | 3 | 3 | * | * | 3.2 | 2.3715 | | |
| ANR | 3 | 3 | 001 | +49 | 7.0 | 2.7127 | 600 °C | 3 | 3 | **141** | **+26** | 22.9 | 2.9126 | 251 | 80 |
| | 3 | 3 | * | * | 3.6 | 2.4486 | 400 Oe | 3 | 3 | * | * | 4.1 | 2.5180 | | |
| SDA | 3 | 3 | 059 | −41 | 30.6 | 2.9347 | 500 °C | 3 | 3 | **054** | **−48** | 20.7 | 2.9036 | 344 | 105 |
| | 3 | 3 | 054 | −46 | 82.9 | 2.9759 | 400 Oe | 3 | 3 | 050 | −50 | 92.4 | 2.9783 | | |
| SDB | 3 | 4 | 060 | −35 | 277.2 | 2.9928 | 500 °C | 3 | 4 | **054** | **−42** | 172.0 | 2.9884 | 341 | 108 |
| | 3 | 3 | 059 | −35 | 1031.4 | 2.9981 | 400 Oe | 3 | 3 | 057 | −40 | 575.0 | 2.9965 | | |
| SDC | 3 | 3 | 095 | −11 | 64.9 | 2.9692 | 500 °C | 3 | 3 | **115** | **−12** | 243.6 | 2.9918 | 342 | 109 |
| | 3 | 3 | 097 | −12 | 41.0 | 2.9512 | 400 Oe | 3 | 3 | 109 | −10 | 127.7 | 2.9843 | | |
| SDD | 1[a] | 1 | 113 | −05 | — | — | 500 °C | 3 | 3 | **142** | **−18** | 240.7 | 2.9917 | 343 | 110 |
| | 3 | 3 | 126 | −10 | 135.3 | 2.9852 | 400 Oe | 3 | 3 | 135 | −10 | 639.0 | 2.9969 | | |
| SDE | 3 | 4 | 090 | −14 | 6.1 | 2.6730 | 500 °C | 3 | 4 | * | * | 2.0 | 1.9878 | 353 | 120 |
| | 3 | 3 | 087 | −24 | 7.1 | 2.7175 | 400 Oe | 3 | 3 | **079** | **+03** | 26.3 | 2.9241 | | |
| SDF | 3 | 3 | 114 | −24 | 262.5 | 2.9924 | 500 °C | 3 | 3 | **131** | **−22** | 315.8 | 2.9937 | 343 | 118 |
| | 3 | 3 | 112 | −22 | 156.7 | 2.9872 | 400 Oe | 3 | 3 | 121 | −24 | 543.6 | 2.9963 | | |
| *Main Section* | | | | | | | | | | | | | | | |
| SBA | 3 | 3 | 009 | −09 | 30.8 | 2.9351 | 500 °C | 3 | 3 | **019** | **−03** | 8.0 | 2.7499 | 287 | 98 |
| | 3 | 3 | 007 | −10 | 28.0 | 2.9284 | 400 Oe | 3 | 3 | 009 | −03 | 13.1 | 2.8478 | | |
| SKA | 3 | 3 | 028 | −09 | 85.0 | 2.9765 | 500 °C | 3 | 3 | **032** | **−13** | 72.3 | 2.9723 | 287 | 99 |
| | 3 | 3 | 029 | −09 | 361.2 | 2.9945 | 400 Oe | 3 | 3 | 031 | −08 | 661.8 | 2.9970 | | |
| SBB | 3 | 3 | 024 | −13 | 12.1 | 2.8343 | 500 °C | 3 | 3 | **032** | **−07** | 18.4 | 2.8910 | 286 | 100 |
| | 3 | 3 | 031 | −09 | 33.4 | 2.9402 | 400 Oe | 3 | 3 | 032 | −05 | 85.5 | 2.9766 | | |
| SKB | 3 | 3 | 038 | −10 | 120.8 | 2.9834 | 500 °C | 3 | 3 | **042** | **−10** | 227.9 | 2.9912 | 286 | 100 |
| | 3 | 3 | 039 | −10 | 53.9 | 2.9629 | 400 Oe | 3 | 3 | 042 | −09 | 71.8 | 2.9722 | | |
| SKZ | 3 | 3 | 016 | −19 | 11.5 | 2.8254 | 500 °C | 3 | 3 | **036** | **+05** | 134.6 | 2.9851 | 286 | 100 |
| | 3 | 3 | 015 | −19 | 8.4 | 2.7630 | 400 Oe | 3 | 3 | 010 | −06 | 30.6 | 2.9346 | | |
| SBC | 3 | 3 | 034 | −03 | 14.4 | 2.8614 | 500 °C | 3 | 3 | **036** | **−06** | 15.5 | 2.8711 | 285 | 90 |
| | 3 | 3 | 038 | −02 | 12.8 | 2.8436 | 400 Oe | 3 | 3 | 039 | −03 | 15.7 | 2.8728 | | |
| SKC | 3 | 4 | 026 | −24 | 130.2 | 2.9846 | 500 °C | 3 | 4 | **032** | **−28** | 143.7 | 2.9861 | 283 | 101 |
| | 3 | 3 | 026 | −25 | 178.1 | 2.9888 | 400 Oe | 3 | 3 | 029 | −25 | 415.5 | 2.9952 | | |
| SBD | 4 | 4 | 025 | −24 | 101.7 | 3.9705 | 500 °C | 4 | 4 | **030** | **−26** | 68.3 | 3.9561 | 286 | 104 |
| | 4 | 4 | 029 | −25 | 75.8 | 3.9604 | 400 Oe | 4 | 4 | 028 | −27 | 70.9 | 3.9577 | | |
| SKD | 3 | 3 | 037 | +04 | 133.0 | 2.9850 | 500 °C | 3 | 3 | **042** | **+05** | 321.8 | 2.9938 | 287 | 96 |
| | 3 | 3 | 037 | +04 | 169.2 | 2.9882 | 400 Oe | 3 | 3 | 039 | +03 | 377.7 | 2.9947 | | |

TABLE 1 (*Continued*)

| | NRM | | | | | | | CLEANED | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | *N* | *S* | *D* | *I* | *k* | *R* | T/H | *N* | *S* | *D* | *I* | *k* | *R* | Strike | Dip |
| SBE | 3 | 3 | 063 | −05 | 5.4 | 2.6293 | 500 °C | 3 | 3 | **078** | **−17** | 5.4 | 2.6310 | 295 | 95 |
| | 3 | 3 | 056 | −07 | 14.5 | 2.8621 | 400 Oe | 3 | 3 | 061 | −13 | 11.2 | 2.8214 | | |
| SKE | 3 | 3 | 054 | −19 | 111.4 | 2.9821 | 500 °C | 3 | 3 | **062** | **−25** | 142.2 | 2.9859 | 290 | 99 |
| | 3 | 3 | 055 | −18 | 67.3 | 2.9703 | 400 Oe | 3 | 3 | 066 | −20 | 45.0 | 2.9555 | | |
| SBF | 3[a] | 3 | 046 | −03 | 54.3 | 2.9631 | 500 °C | 4 | 4 | **054** | **−04** | 102.5 | 3.9707 | 290 | 94 |
| | 4 | 4 | 046 | −00 | 100.5 | 3.9701 | 400 Oe | 4 | 4 | 048 | +01 | 68.6 | 3.9562 | | |
| SKF | 3 | 3 | 030 | +08 | 82.0 | 2.9756 | 500 °C | 3 | 3 | **040** | **+18** | 24.8 | 2.9193 | 294 | 94 |
| | 3 | 3 | 035 | +07 | 43.9 | 2.9545 | 400 Oe | 3 | 3 | 034 | +16 | 58.0 | 2.9655 | | |
| SBG | 3 | 3 | 059 | −25 | 93.9 | 2.9787 | 500 °C | 3[b] | 3 | **061** | **−30** | 28.3 | 2.9293 | 291 | 93 |
| | 3 | 3 | 053 | −22 | 58.7 | 2.9659 | 400 Oe | 3 | 3 | 056 | −24 | 29.7 | 2.9326 | | |
| SKG | 3 | 3 | 024 | −01 | 4378.2 | 2.9995 | 500 °C | 3 | 3 | **026** | **−07** | 244.9 | 2.9918 | 290 | 93 |
| | 3 | 3 | 023 | −02 | 1270.2 | 2.9984 | 400 Oe | 3 | 3 | 026 | −02 | 172.7 | 2.9884 | | |
| SBH | 3 | 3 | 027 | −21 | 194.6 | 2.9897 | 500 °C | 3 | 3 | **030** | **−21** | 596.1 | 2.9966 | 288 | 94 |
| | 3 | 3 | 025 | −23 | 197.2 | 2.9899 | 400 Oe | 3 | 3 | 030 | −21 | 166.2 | 2.9880 | | |
| SKH | 3 | 3 | 356 | −18 | 149.9 | 2.9867 | 500 °C | 3 | 3 | **354** | **−25** | 152.1 | 2.9869 | 288 | 91 |
| | 3 | 3 | 357 | −18 | 121.3 | 2.9835 | 400 Oe | 3 | 3 | 355 | −22 | 169.0 | 2.9882 | | |
| AJA | 2 | 2 | 006 | −19 | 43.5 | 1.9770 | 500 °C | 2 | 2 | **009** | **−27** | 97.1 | 1.9897 | 288 | 92 |
| AJB | 2 | 2 | 032 | +02 | 352.5 | 1.9972 | 500 °C | 2 | 2 | **037** | **00** | 504.9 | 1.9980 | 287 | 92 |
| AJC | 2 | 2 | 022 | +02 | 159.8 | 1.9937 | 500 °C | 2 | 2 | **026** | **−03** | 260.0 | 1.9962 | 286 | 93 |
| AJD | 2 | 2 | 018 | +27 | 12.5 | 1.9202 | 500 °C | 2 | 2 | **025** | **+44** | 6.1 | 1.8348 | 285 | 93 |
| SBI | 3 | 3 | 309 | +79 | 17.0 | 2.8821 | 500 °C | 3 | 3 | **215** | **+66** | 257.6 | 2.9922 | 285 | 94 |
| | 3 | 3 | 252 | +77 | 38.9 | 2.9486 | 400 Oe | 3 | 3 | 226 | +70 | 861.0 | 2.9977 | | |
| AJE | 2 | 2 | 035 | +62 | 15.1 | 1.9337 | 500 °C | 2 | 2 | **170** | **+64** | 28.5 | 1.9649 | 284 | 93 |
| AJF | 2 | 2 | 029 | +73 | 47.9 | 1.9791 | 500 °C | 2 | 2 | **186** | **+70** | 11.4 | 1.9121 | 283 | 90 |
| SKI | 3 | 3 | 169 | +61 | 15.3 | 2.8693 | 500 °C | 3 | 3 | **180** | **+47** | 38.6 | 2.9482 | 283 | 90 |
| | 3 | 3 | 177 | +63 | 27.3 | 2.9268 | 400 Oe | 3 | 3 | 186 | +54 | 34.1 | 2.9413 | | |
| AJG | 2 | 2 | 223 | +30 | 62.8 | 1.9841 | 500 °C | 2 | 2 | **217** | **+25** | 145.9 | 1.9931 | 283 | 90 |
| AJH | 2 | 2 | 296 | +43 | 40.9 | 1.9756 | 500 °C | 2 | 2 | **257** | **+32** | 58.2 | 1.9828 | 281 | 90 |
| SBJ | 3 | 3 | 264 | +20 | 1001.0 | 2.9980 | 500 °C | 3 | 3 | **245** | **+19** | 319.3 | 2.9937 | 281 | 90 |
| | 3 | 3 | 262 | +20 | 42.8 | 2.9533 | 400 Oe | 3 | 3 | 250 | +24 | 163.5 | 2.9878 | | |
| SKJ | 3 | 3 | 223 | +05 | 1283.4 | 2.9984 | 500 °C | 3 | 3 | **214** | **+04** | 418.3 | 2.9952 | 285 | 90 |
| | 3 | 3 | 228 | +05 | 131.3 | 2.9848 | 400 Oe | 3 | 3 | 219 | +03 | 364.6 | 2.9945 | | |

TABLE 1 (*Concluded*)

| Site | NRM | | | | | | | CLEANED | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *N* | *S* | *D* | *I* | *k* | *R* | T/H | *N* | *S* | *D* | *I* | *k* | *R* | Strike | Dip |
| SBK | 4 | 4 | 223 | +19 | 48.1 | 3.9376 | 500 °C | 4 | 4 | **221** | **+14** | 124.7 | 3.9759 | 281 | 97 |
| | 4 | 4 | 223 | +23 | 37.6 | 3.9201 | 400 Oe | 4 | 4 | 221 | +19 | 99.0 | 3.9697 | | |
| SKK | 3 | 3 | 222 | +02 | 51.3 | 2.9610 | 500 °C | 3 | 3 | **216** | **+08** | 73.8 | 2.9729 | 269 | 109 |
| | 3 | 3 | 220 | +01 | 58.6 | 2.9659 | 400 Oe | 3 | 3 | 215 | +08 | 389.1 | 2.9949 | | |
| SBL | 3 | 3 | 225 | +10 | 92.7 | 2.9784 | 500 °C | 3 | 3 | **220** | **+11** | 128.6 | 2.9844 | 275 | 121 |
| | 3 | 3 | 225 | +08 | 134.0 | 2.9851 | 400 Oe | 3 | 3 | 221 | +10 | 391.6 | 2.9949 | | |
| SKL | 3 | 3 | 190 | +02 | 37.2 | 2.9463 | 500 °C | 3 | 3 | **183** | **+13** | 215.5 | 2.9907 | 274 | 120 |
| | 3 | 3 | 183 | +05 | 73.0 | 2.9726 | 400 Oe | 3 | 3 | 184 | +15 | 214.3 | 2.9907 | | |
| SBM | 3 | 3 | * | * | 3.9 | 2.4892 | 500 °C | 3 | 3 | **200** | **+27** | 15.5 | 2.8709 | 278 | 106 |
| | 3 | 3 | 247 | +37 | 47.0 | 2.9574 | 400 Oe | 3 | 3 | 227 | +38 | 34.5 | 2.9421 | | |
| SCA | 3 | 3 | 042 | +66 | 33.8 | 2.9408 | 500 °C | 3 | 3 | **080** | **+81** | 41.0 | 2.9513 | 290 | 110 |
| SCB | 3 | 3 | 193 | +13 | 27.1 | 2.9262 | 500 °C | 3 | 3 | **188** | **+11** | 71.6 | 2.9721 | 286 | 107 |
| SCC | 3 | 3 | 194 | +35 | 25.7 | 2.9223 | 500 °C | 3 | 3 | **194** | **+24** | 101.5 | 2.9803 | 286 | 106 |
| SCD | 3 | 3 | 194 | 00 | 23.4 | 2.9146 | 500 °C | 3 | 3 | **189** | **+02** | 874.9 | 2.9977 | 275 | 105 |
| SCE | 3 | 3 | 195 | +19 | 44.3 | 2.9548 | 500 °C | 3 | 3 | **184** | **+18** | 462.4 | 2.9957 | 280 | 109 |
| SBN | 3 | 5 | * | * | 1.6 | 1.7739 | 500 °C | 3 | 5 | **226** | **−03** | 63.4 | 2.9685 | 293 | 112 |
| | 3 | 3 | * | * | 2.6 | 2.2340 | 400 Oe | 3 | 3 | 229 | −02 | 17.7 | 2.8869 | | |
| SKN | 3 | 3 | 183 | +38 | 24.2 | 2.9175 | 500 °C | 3 | 3 | **184** | **+35** | 459.5 | 2.9956 | 286 | 120 |
| | 3 | 3 | 173 | +51 | 43.5 | 2.9540 | 400 Oe | 3 | 3 | 184 | +37 | 424.4 | 2.9953 | | |
| SBO | 3 | 3 | 320 | +12 | 15.2 | 2.8687 | 500 °C | 3 | 3 | **216** | **+16** | 25.5 | 2.9216 | 276 | 112 |
| | 3 | 3 | 232 | +09 | 9.5 | 2.7893 | 400 Oe | 3 | 3 | 224 | +13 | 22.5 | 2.9113 | | |
| SBP | 3 | 3 | 239 | +50 | 12.8 | 2.8437 | 500 °C | 3 | 3 | **225** | **+47** | 17.0 | 2.8823 | 289 | 124 |
| | 3 | 3 | 237 | +50 | 9.7 | 2.7936 | 400 Oe | 3 | 3 | 228 | +49 | 13.1 | 2.8477 | | |
| SKP | 3 | 3 | 251 | +19 | 48.1 | 2.9585 | 500 °C | 3 | 3 | **241** | **+19** | 87.4 | 2.9771 | 294 | 115 |
| | 3 | 3 | 256 | +17 | 99.6 | 2.9799 | 400 Oe | 3 | 3 | 249 | +20 | 79.6 | 2.9749 | | |
| SBQ | 3 | 3 | 268 | −39 | 26.4 | 2.9241 | 450 °C | 3 | 3 | **200** | **+13** | 7.4 | 2.7307 | 269 | 108 |
| | 3 | 3 | * | * | 3.7 | 2.4526 | 1000 Oe | 3 | 3 | 224 | −04 | 33.9 | 2.9410 | | |

$N$ = number of samples; $S$ = number of specimens; $D$, $I$ = site mean declination, inclination; $k = N - 1/N - R$.
$R$ = length of sum of $N$ unit vectors; Strike, Dip = geological strike and dip; strike is measured East of true North and dip is to the right when looking along strike. Dips > 90°indicate overturning (way-upness confirmed by cross bedding).
[a]Some NRM data missing (SBF, SDD, SEB).
[b]One specimen previously AF treated to 1800 Oe (SBG).
[c]Insufficient material for both AF and thermal full statistics (SEB).
*Directions random (SBM, SBN, SBQ, SEA, SEB, ANR, SDE).
Data in boldface type is used in further analysis.

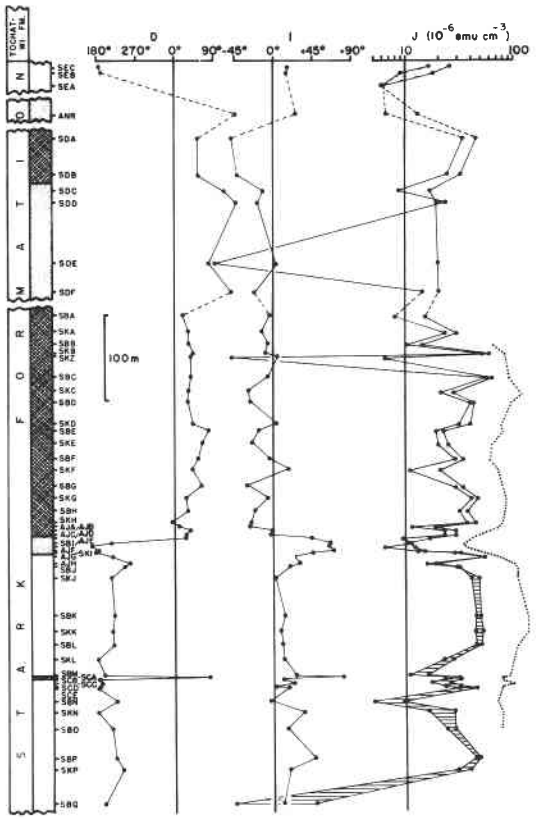



FIG. 2. Site mean paleomagnetic data, displayed stratigraphically. Site names are indicated. Left column shows geologic formation. The second column displays polarity implied by the paleomagnetic data: crosshatched, reversed; stippled, intermediate; white, normal polarity. *D* and *I* are declination east of true north and inclination below the horizontal of cleaned site mean paleomagnetic directions (bold faced data in Table 1). J is site mean intensity of magnetization before and after cleaning; shading between these curves indicates that the cleaned intensity is greater than the NRM intensity. The dotted intensity curve is 5 point running mean of the cleaned intensities shifted right for clarity by a factor of 3 (omitting sites SBQ and SKZ).

mately half way through the main section, that part of the section stratigraphically above site AJD possessing 'reversed' directions and that part below site SKI possessing 'normal' directions; sites between these two groups have intermediate directions (Fig. 2). 'Normal' and 'reversed' are used in the sense defined by Reid (1972); namely, that polarity which yields north (south) poles lying on the polar wander curve traceable to the present north pole is termed normal (reversed). In what follows, it is often convenient to discuss the two polarity groups separately.

*Alternating Field Demagnetization*

Detailed step-wise AF demagnetization studies up to 1800 Oe peak field were carried out on pilot specimens from 32 of the 45 sites. With few exceptions, the behaviour of all specimens is both remarkably uniform and very stable.

The magnetization vectors of specimens from all but 6 of the 32 pilot specimens moved less than 15° between NRM and 400 Oe. Between 400 and 1800 Oe, very little further angular change was observed; in only two cases did movement exceed 8° (SBM, 14°; SCE, 32°).

On the basis of the uniformity of behaviour throughout the section, 400 Oe has been chosen as that AF treatment which best removes secondary components without unduly degrading the primary remanence. However, it is clear from the stability of the remanence directions throughout AF cleaning that this choice is not critical. One specimen from each hand block from all SB- and SK- sites (32 sites, 99 samples) was therefore treated at 400 Oe peak field.

Under AF demagnetization the intensities of magnetization follow two distinct patterns, one for reversed specimens and one for normal specimens. All but one of the pilot specimens from the reversed sites show a *decrease* in $J/J_0$ (to between 77 and 96% at 400 Oe) (Fig. 3a). The aberrant site, SKZ, is green and is magnetically much softer. On the other hand, all but three of the normal sites show an initial intensity *increase* (100 to 172% at 400 Oe) (Fig. 3b). Two of the three exceptions, sites SBN and SCE, possess anomalously large, soft components that are removed by 100 to 200 Oe, and thereafter they behave like the others (SCE, being somewhat greener than the others, is very weakly magnetized, and thus $J/J_0$ appears to vary more erratically). The remaining site, SBQ, is lithologically different and will be discussed in a later subsection.

The observed pattern of intensity behaviour is readily explicable in terms of small secondary components of magnetization parallel to the present field at the collecting locality. In order to facilitate discussion of common features, we return to this point below, after results from other stratigraphic sections have been discussed.

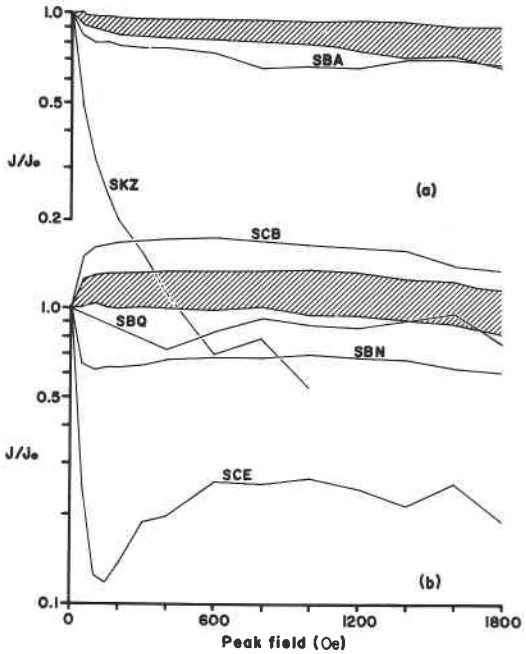



FIG. 3. Normalized intensity of main section pilot specimens under AF treatment. (a) Reversed sites: hatched area includes specimens from sites SBB, SBC, SBD, SBE, SBF, SBG, SBH, SKA, SKB, SKC, SKD, SKF, and SKH. (b) Normal sites: hatched area includes specimens from sites SBI, SBJ, SBK, SBL, SBM, SBO, SBP, SCA, SCD, SKI, SKJ, SKK, and SKP.

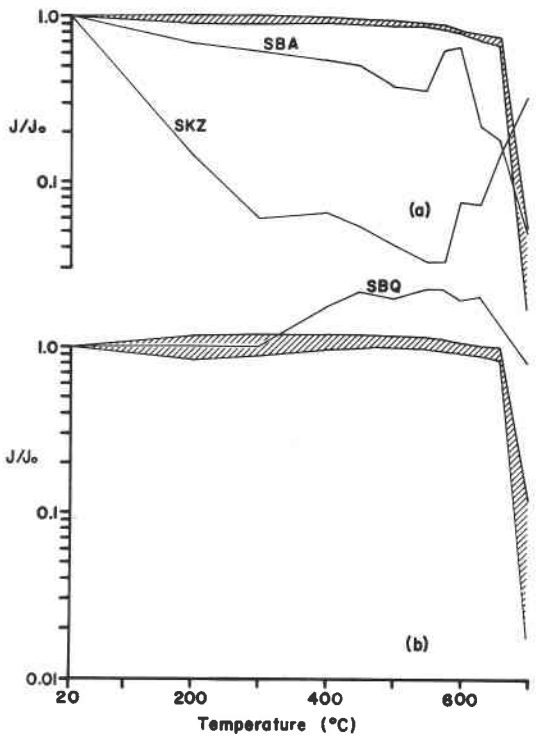



FIG. 4. Normalized intensity of main section pilot specimens under thermal treatment. (a) Reversed sites: hatched area includes specimens from sites SBD, SBF, and SBH. (b) Normal sites: hatched area includes specimens from sites SBJ, SBL, SBN, SBP, and SCE.

*Thermal Demagnetization*

Uniformity of the lithology and magnetic behaviour under AF treatment led us to restrict detailed thermal demagnetization studies to pilot specimens from 11 sites, spread throughout the main section. Three of these sites are lithologically distinct (SBA, SKZ, SBQ) and behave differently; they will be discussed separately in the next subsection. Between NRM and 500 °C, six of the remaining eight moved less than 20°, while the other two moved 27° and 55°. From 500 to 660 °C, they all moved less than 8°; but above 660 °C, movements were large.

As with the AF treatment, the exact choice of temperature for blanket cleaning is not critical; 500 °C was chosen, although any temperature between 300 and 550 °C would probably produce statistically indistinguishable results when applied to the section as a whole, for the endpoints are generally well defined within that interval.

The behaviour of $J/J_o$ under heating for these 8 sites was remarkably uniform, and the same contrast in behaviour between normal and reversed sites as observed under AF treatment is apparent (Fig. 4). The intensities in the reversed part of the section decrease slightly, by 500 °C ($J/J_o$ between 85% and 100%), dropping more steeply towards 660 °C, before falling sharply one to nearly two orders of magnitude by 700 °C as the hematite Curie temperature is exceeded (Fig. 4a). In the normal part of the section, there is an initial increase up to 500 °C ($J/J_o$ between 100% and 120%); thereafter, the behaviour is the same as for the reversed specimens (Fig. 4b).

*Sites SBA, SKZ, SBQ*

These three sites from the main section are discussed separately because they possess distinct characteristics not adequately described by the general discussion above.

The behaviour of SBA pilot specimens under

AF and thermal cleaning is the same as other specimens with the exception of slightly steeper initial drops in $J/J_0$ (Figs. 3a and 4a). However, above 500 °C behaviour is anomalous, showing large angular movement and a sharp increase in $J/J_0$ between 550 and 630 °C (Fig. 4a). It appears that new magnetic material is being created at higher temperatures. However, good end points are reached by 400 Oe and 500 °C; therefore, the same blanket treatments as for the other sites have been applied.

Site SKZ was collected in a green bed and all specimens were weakly magnetized ($J_0 \sim 5 \times 10^{-6}$ emu/cm$^3$) and magnetically soft, making measurements difficult. Detailed AF and thermal cleaning were carried out on one specimen from each block. Though $J/J_0$ for SKZ specimens drops very much more rapidly under both AF and thermal treatment than for the rest of the section (Figs. 3a and 4a), good end points are reached by all specimens between 300 and 400 Oe and between 400 and 500 °C. Above 500 °C, the behaviour becomes somewhat erratic. Since the 'standard' AF and thermal treatments of 400 Oe and 500 °C applied to the rest of the section fall within these end point ranges, these treatments have been selected for further analysis. It is noteworthy that, despite the green coloration, the calculated site mean does not differ greatly from those of nearby red sites.

Site SBQ, at the very bottom of the main section, consists of pink dolomite that is extremely weakly magnetized ($J_0 \sim 1 \times 10^{-7}$ emu/cm$^3$). Under AF treatment, specimens proved to be magnetically very hard and good end points were reached between 800 and 1200 Oe. The result after 1000 Oe treatment has been chosen as representative. During thermal cleaning, end points were reached by two specimens between 400 and 600 °C; the third specimen's end point occurred between 200 and 450 °C. Thus, the 450 °C data are used in further analysis.

*Other Stratigraphic Sections*

Besides the 45 sites in the main section, samples were collected at 10 additional sites in 3 localities. The SD- and SE- sections (6 and 3 sites, respectively) are located about 3 km north of the main section and a single site, ANR, lies 14 km to the east (Fig. 1).

While the final magnetic directions of sites in the SD- and SE- sections differ considerably from those of the main section, in their response

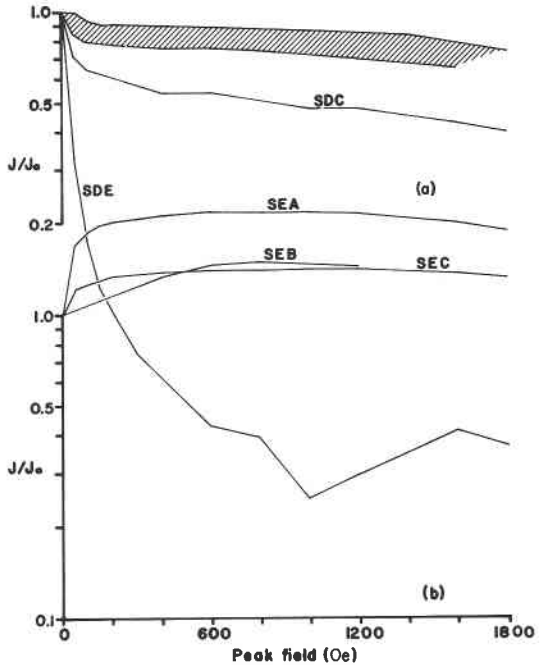



FIG. 5. Normalized intensity of pilot specimens under AF treatment. (a) SD- section sites: hatched area includes specimens from sites SDA, SDB, SDD, and SDF. (b) SE- section sites.

to cleaning these sections are similar respectively to the reversed and normal parts of the main section and have therefore been treated in parallel manner (compare Figs. 5 and 6 with Figs. 3 and 4). An exception to this behaviour is green site SDE, the magnetic intensity of which drops much more steeply than the rest of the sites under both thermal and AF treatment (Figs. 5a and 6a).

For all sites, good thermal end points enabled the 'standard' 500 °C treatment to be applied as previously to one specimen from each block. AF end points bracketing 400 Oe were reached by all sites except SEA and SEB, allowing the same 400 Oe blanket treatment used for the main section to be applied. The latter two sites exhibited end points in the range 600 to 1200 Oe, and 800 Oe treatment has been selected as representative.

Specimens from site ANR cleaned by AF and thermal treatment behave similarly to those from the normal part of the main section. Though initial angular movement is large (60 to 70°), good end points were reached between 400 and 800 Oe; so, 400 Oe has been chosen as an

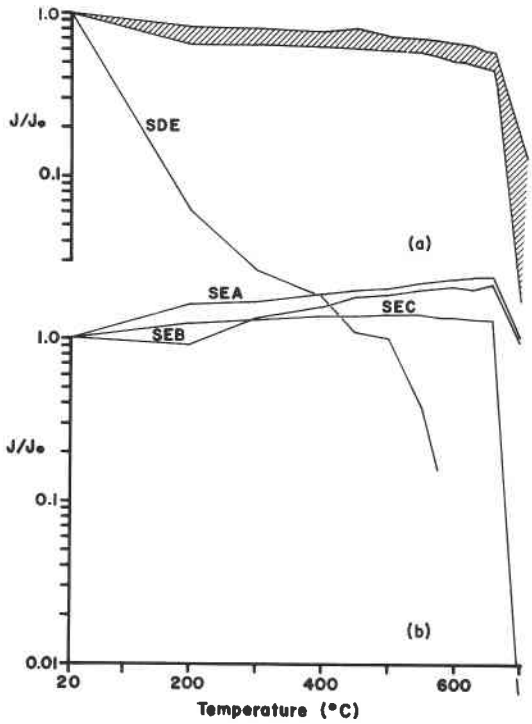



FIG. 6. Normalized intensity of pilot specimens under thermal treatment. (a) SD- section sites: hatched area includes specimens from sites SDA, SDB, SDC, SDD, and SDF. (b) SE- section sites.

appropriate AF treatment. One specimen from each of the three blocks was thermally treated. Thermal end points were reached by two of the specimens between 500 and 630 °C; the third specimen's end point occurred between 600 and 630 °C. Therefore, the 600 °C data have been selected for further analysis.

*Summary of Site Mean Directions and Intensities*

Following pilot demagnetization studies, at least one specimen from each hand block was treated at 400 Oe peak alternating field and 500 °C, with the exceptions noted in Table 1. In 12 cases, site means are statistically random (Watson 1956*a*; Vincenz and Bruckshaw 1960); 8 of these are NRM's, 2 are thermal results, and 2 represent magnetically cleaned data. For one site (SEA), both thermal and AF results are random, causing this site to be rejected from further consideration.

Within-site scatter is generally similar for both thermal and AF treatments and the site mean directions obtained by both methods are in good agreement. For the 37 sites for which appropriate data exist, the angular difference between thermal and AF results has an average of 7.5°. This indicates that conclusions based on either set of data separately would be essentially identical. For convenience, therefore, we restrict further discussion to the thermal results (bold face type in Table 1), in accord with most studies on hematite-bearing red beds. However, the overall result obtained from magnetic cleaning is given in Table 2 (entry 6) for comparison. There is one exception—site SDE,

TABLE 2. Stark Formation mean directions and poles

| | Directions | | | | | | Poles | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $N$ | $D$ | $I$ | $\alpha_{95}$ | $k$ | $R$ | $\lambda$ | $\phi$ | dp | dm |
| 1. Corrected for geological dip | 39 | 213.6 | +15.2 | 6.7 | 12.6 | 35.9959 | 15.3° S | 145.3° W | 3.5 | 6.9 |
| 2. Before dip correction | 39 | 272.4 | −71.5 | 6.6 | 12.9 | 36.0557 | 46.5° S | 123.2° E | 10.1 | 11.6 |
| 3. Corrected for plunge and geological dip | 39 | 218.7 | +14.3 | (as entry 1) | | | 14.3° S | 150.5° W | (as entry 1) | |
| 4. Reversed sites, corrected for geological dip | 20 | 035.9 | −11.7 | 9.1 | 13.7 | 18.6142 | 16.4° N | 031.9° E | 4.7 | 9.2 |
| 5. Normal sites, corrected for geological dip | 19 | 211.1 | +18.7 | 10.2 | 11.8 | 17.4790 | 14.1° S | 142.4° W | 5.5 | 10.6 |
| 6. Corrected for geological dip | 30 | 216.6 | +13.8 | 7.6 | 12.9 | 27.7468 | 15.1° S | 148.5° W | 4.0 | 7.8 |

$N$ = number of sites; $D$, $I$, $k$, $R$ as in Table 1; $\alpha_{95}$ = radius of 95% circle of confidence in degrees (Fisher 1953); $\lambda$, $\phi$ = latitude, longitude of pole; dp, dm = semi-major, semi-minor axes of 95% oval of confidence.
All entries refer to the 'main' section only. Entries 1–5 refer to thermally cleaned data, entry 6 to AF cleaning. In groups including both polarities, reversed sites have been inverted (entries 1, 2, 3, and 6).

where randomness of the thermal result forced use of the 400 Oe data.

Irving *et al.* (1966) and Tarling (1966) have found that remanent intensities of sediments as well as volcanics are log normally distributed. Therefore, geometric mean remanent intensities have been calculated following the same hierarchy as in finding mean directions.

Except for the non-red sites SBQ and SKZ, intensities throughout the main sections vary by about an order of magnitude from about 5 to $50 \times 10^{-6}$ emu/cm$^3$. However, the majority lie in the range 10 to $40 \times 10^{-6}$ emu/cm$^3$. The contrast already noted between normal and reversed samples is also clear at the site level, the intensity curves for NRM and for cleaned results crossing in the vicinity of the polarity transition zone (Fig. 2).

## Discussion

### *Secondary Magnetization Removed by Partial Demagnetization*

The remarkable pattern of intensity behaviour during AF cleaning suggests that a secondary component of magnetization, approximately parallel to the magnetization of the reversed sites but opposing the normal sites, is being removed by fields up to 400 Oe. A recently acquired remanence directed along the present geomagnetic field at the collecting locality and with the strata in their present attitude is an obvious candidate. This suggestion has been tested by analyzing the magnetic vectors removed during 'cleaning'. Of the 55 sites sampled, 42 have been subjected to AF treatment. The vectors removed between NRM and 400 Oe form a coherent group, the mean of which lies close to the present field. Details are as follows. Site level statistics on the removed vectors indicate that at 8 sites, these are random (Watson 1956*a*). Considering the 34 non-random sites, it may be preferable to weight in favour of those with higher within-site precision. Seventeen of the 34 have within-site precision parameters greater than 10, and their mean direction is $D = 013$, $I = +83$ ($k = 17$, $\alpha_{95} = 9°$); inclusion of the 17 sites which have $k < 10$ moves the mean only 4° and yields $D = 347$, $I = +87$ ($k = 18$, $\alpha_{95} = 6°$). Neither of these directions differ significantly from the present field ($D = 030$, $I = +83$) at the collecting locality.

The 34 sites used in this analysis represent both normal and reversed polarities and transitional directions, but it is clear that vectorial subtraction succeeds in isolating a secondary magnetization common to the various groups. Application of Graham's (1949) fold test indicates that the removed vectors are significantly more scattered after restoring the strata to the horizontal, thereby demonstrating that they were imposed after folding. For the 17 sites having within-site $k > 10$, we obtain $k = 8$ (*cf.* $k = 17$ before structural correction); whereas for all 34 sites, we obtain $k = 7$ (*cf.* $k = 18$ before correction). After structural correction, the 17 sites with $k > 10$ are significantly more scattered at the 95% confidence level, whereas all 34 sites are more scattered at the 99% level (McElhinny 1964).

The negative fold test and close proximity of the overall mean for the removed vectors with the present field at the sampling locality leave little doubt that most specimens possess a secondary magnetization acquired in very recent times. However, demagnetization to 400 Oe successfully erases this magnetic component and reveals an underlying magnetization whose nature is discussed further below.

A similar analysis of the vectors removed by heating to 500 °C has been carried out. Again a coherent grouping emerges with 51 non-random sites yielding a mean direction of $D = 239$, $I = +80$ ($k = 22$, $\alpha_{95} = 4°$). After correction for local geological dip, scatter increases significantly at the 99% confidence level ($k$ decreases to 6), and a post-folding secondary component is indicated. However, the mean of these removed vectors lies 17° from the present field and 12° from the mean of the 34 magnetically treated sites described above. It is not clear at the present time whether this small offset has any geophysical significance.

### *Formation mean direction of magnetism*

Sites SKI through AJD have intermediate directions representing the transition between the normal and reversed parts of the main section. These sites are therefore excluded from the calculation of the overall mean direction. Also excluded from the main section is site SCA, which is interpreted as a geomagnetic field excursion. Site mean directions of the remaining normal and reversed sites are shown in Fig. 7. A mean direction of $D = 214°$, $I = +15°$

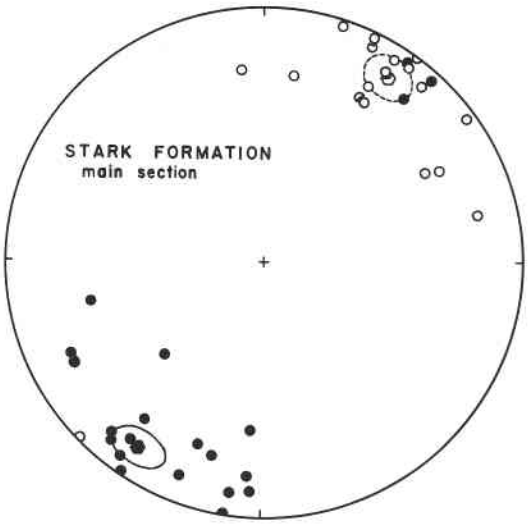



FIG. 7. Site mean directions from the main section, normal and reversed sites only. Hexagons represent both ends of the section mean direction (entry 1, Table 2), with $\alpha_{95}$ circles of confidence. Open (closed) symbols indicate upper (lower) hemisphere. Equal area projection.

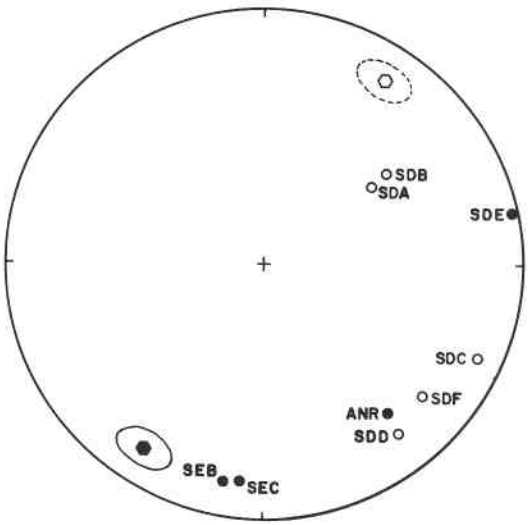



FIG. 8. Site mean directions from the SD- and SE- sections and site ANR. Symbols as in Fig. 7.

(entry 1, Table 2) for that part of the Stark Formation represented by the main section is obtained by giving unit weight to all 39 non-intermediate sites and inverting reversed sites.

Site mean directions for sites of the SD- and SE- sections and site ANR are shown in Fig. 8. Four of the nine sites involved agree reasonably well with the main section results; these are sites SEB and SEC, which lie within the normal group of the main section, and sites SDA and SDB, which are marginal to the reversed group. Combining these four site means with the 39 main section site means moves the overall mean by less than 2°.

All the results quoted thus far were computed under the assumption, commonly followed in paleomagnetic studies, that the folds involved do not plunge. Since most of the strata under study here possess steep dips, the possibility of plunge represents a serious potential problem. To test the possible significance of plunge, an analysis of bedding plane poles has been carried out following the method of Watson (1960), outlined by Cruden (1968).

The data used consisted of 62 bed orientations measured in the field during paleomagnetic sampling and strikes and dips measured from Barnes' (1951) map. The 26 strikes and dips from Barnes' map are more or less randomly distributed over the area included in the left half of Fig. 1. The 36 bed attitudes measured in the field are distributed along the coast line between the main and SE- sections (the 45 closely spaced main section attitudes have been combined into a single datum to avoid over-representation) and on the south arm of Pearson Point (just off the north edge of Fig. 1). Because it is not known exactly where Barnes measured his bed attitudes, it is possible that there is some duplication within this set of data; however, such doubling of points appears to be limited to a maximum of 5 points. Excluded from the analysis are two small areas that give widely scattered bedding poles and thus appear to represent tectonically disturbed areas. No samples were collected from one of the areas, but the second area covers the East half of the island where the SD section lies (Fig. 1). Although this part of the island appears to be tectonically complex, the six sites of the SD section themselves represent a homoclinal sequence (Table 1), and this is important to the interpretation of their paleomagnetic data (see below).

The 62 bedding poles yield the eigenvector corresponding to the minimum eigenvalue of the variance–covariance matrix of direction cosines (Cruden 1968) and the $\pi$-diagram shown in Fig. 9. The distribution of points clearly indicates approximately cylindrical folding. The eigenvector associated with the minimum eigen-

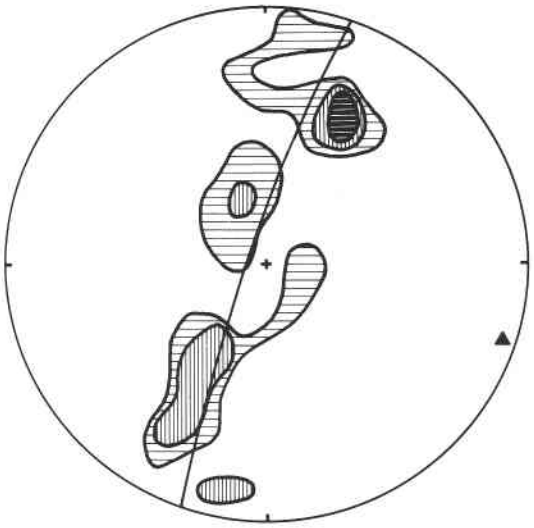


FIG. 9. Equal area $\pi$-diagram showing the distribution of 62 bedding poles in the sampled region. Contours at 2, 5, and 10% of poles per 1% area. Triangle indicates the location of the eigenvector corresponding to the minimum eigenvalue, which gives the axis of folding.

value, which corresponds to the fold axis, lies at 110°, +5°. There are insufficient data available for rigorous application of the statistical tests described by Cruden. However, it is clear that the plunge cannot amount to more than a few degrees. It is noteworthy that if the same methodology is applied to the 45 closely spaced bed attitudes from the main section, which represent a homoclinal sequence on one limb of a fold, a plunge axis is obtained that is only 5° removed from that arrived at above; however, due to the tighter distribution of bedding poles ($k = 51$), the uncertainty in this latter determination is correspondingly greater.

The plunge found above was used to correct the main section paleomagnetic mean and the result is not significantly different from the no-plunge result (compare entries 1 and 3, Table 2). Such a small difference will only become noteworthy if and when very much more data become available; and therefore, entry 1 of Table 2 is used henceforth.

*Geomagnetic Field Behaviour*

The proposed stratigraphic relationships between the paleomagnetic sampling sections are illustrated in Fig. 2. The stratigraphic positions of the SE section and site ANR are obtained by direct field measurement to the base of the overlying Tochatwi Formation. This arrangement assumes that the average sedimentation rates at the two locations separated by 14 km did not differ by more than a factor of 2.

Hoffman (1968, p. 36) describes "two distinctive stromatolitic carbonate beds" having thickness of about 85 and 125 ft (about 26 and 38 m) in the upper part of the Stark Formation. He concludes that both these beds lie above the strata we have called the main section. On the other hand, the SD section appears from field evidence to be stratigraphically above at least one of these two carbonate beds. These geological constraints are all satisfied by the scheme adopted in Fig. 2. Hoffman (1968) ascribes a minimum thickness of about 2000 ft (610 m) to the Stark Formation. The sections sampled add up to 840 m (Fig. 2), of which the main section itself represents 600 m. Thus, we have exceeded the minimum thickness and in all probability our sampling includes most of the entire Stark Formation.

It appears from the paleomagnetic data that there is essentially no temporal overlap between the different sections, and the interpretation given (Fig. 2) leads to the simplest sequence of geomagnetic polarities. Thus, there appears to have been two full field reversals during Stark times: one recorded in the middle of the main section and one somewhere between the SD and SE sections. It is perhaps worth pointing out that the polarity pattern emerging has considerable potential in regional and possibly global correlation, particularly in sedimentary sequences devoid of index fossils and sparsely endowed with datable igneous material.

The virtual geomagnetic poles (VGP's) corresponding to the intermediate sites (SKI through AJD) follow an approximate great circle path (Fig. 10), as has been found for studies of more recent polarity transitions (*e.g.* Creer and Ispir 1970*a*; Dunn *et al.* 1971). Furthermore, strata in the transition zone are less strongly magnetized than the rocks above and below. This point is more clearly revealed when five point running (geometric) means of the thermal results (excluding sites SKZ and SBQ, which are lithologically distinct) are computed (Fig. 2). While it is recognized that the relationship between measured remanent intensity and paleointensity of the geomagnetic field is not necessarily simple, the marked lithological uniformity of the section makes it not unlikely that the drop in intensity recorded during the

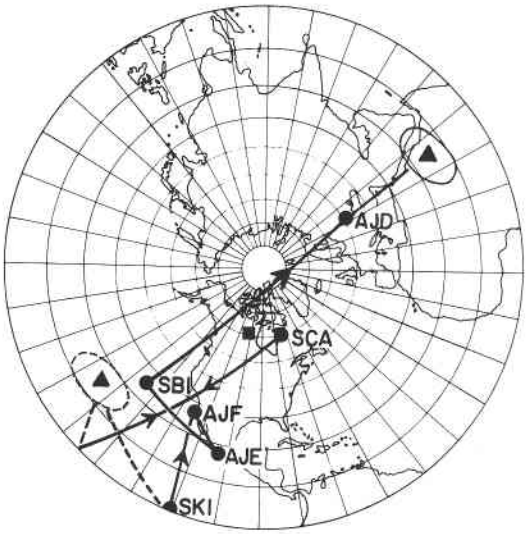



FIG. 10. Virtual geomagnetic poles for the field excursion (site SCA) and for the sites representing the main section polarity transition. Closed (open) symbols and solid (dashed) lines are on the North (South) hemisphere. Triangles represent the overall Stark Formation paleomagnetic pole (entry 1, Table 2). Square is the sampling location.

transition represents a real drop in paleointensity. A decrease in paleointensity during more recent reversals by a factor of about 4 or 5 has been noted by several workers, as reviewed by McElhinny (1971). The decrease observed in the Stark Formation represents a factor of about 2 or 3. This feature, along with the path of transitional poles, indicates that in some ways this reversal closely resembles those recorded in much more recent rocks.

The polarity transition occupies approximately 25 m of section (Fig. 2), and it therefore contrasts markedly with results from several Phanerozoic reversals, which occupy only 1 or 2 m of red beds (see Baag and Helsley 1974, for a recent review), and sometimes as little as a few centimetres (McElhinny 1970). Thus it is tempting to suggest that our results favour a slower reversal mechanism in the early Proterozoic, but a correspondingly rapid sedimentation rate cannot be ruled out. Of course, it is important to distinguish between the rate at which the sediments in question were deposited and the rate at which they acquired their remanence, since the two are not necessarily the same. We return to this point below, when the origin of the magnetization is discussed. If we accept, for the moment, that magnetization occurred at or shortly after deposition, and also that the time required for a field reversal was similar to that involved in geologically recent reversals (*i.e.* $1–2 \times 10^3$ years, McElhinny 1971), then a sedimentation rate of about 1–2 cm/y is implied. Although this is rather high compared to observed contemporary rates of accumulation (Kukal 1971), it cannot be ruled out, particularly if a deltaic environment be envisaged. However 1 or 2 cm/y is likely to represent the upper limit, and this being so, the entire sampled sequence (>800 m, see Fig. 2) represents at least $4 \times 10^4$ years, and perhaps as much as $8 \times 10^6$ years.

Site SCA lies roughly in the middle of the normal part of the main section and has a VGP about 90° from the section mean (Fig. 10). This solitary intermediate direction is interpreted as a field excursion event similar to those observed in the more recent geologic record (*e.g.* Doell and Cox 1972; Barbetti and McElhinny 1972). It is noteworthy that this VGP lies close to the path subsequently traced out by the subsequent transition zone. This proximity to the later reversal path may be entirely coincidental, or it may be due to a persistent non-dipole source such as that hypothesized by Watkins and Nougier (1973) to explain the similarity of two separate Bruhnes epoch field excursions recorded in rocks from the Indian Ocean; alternatively, it may be a consequence of the three dipole model proposed by Creer and Ispir (1970*b*) to explain parallism of pole paths during several different polarity transitions.

As pointed out above, the SD section lies in a tectonically disturbed zone. However, the scatter of the six site means cannot be attributed to structure because the entire section forms a homoclinal sequence (Table 1). The results are therefore interpreted as representing a period of geomagnetic instability consisting of a possible excursion (4 coplanar sites, SDF–SDC), followed by a brief attempt to return to reverse polarity (SDB-SDA), before swinging through the ANR direction to the normal polarity of SEB and SEC (Fig. 8). It is worth noting that although structural complexity cannot be invoked as a cause for the scatter of the results, it may explain the offset between the SDB-SDA data and the main section reversed directions (Fig. 8).

*Origin of the Magnetization Revealed by Partial Demagnetization*

The AF and thermal demagnetization experiments indicate that, except for predominantly green sites, the 'cleaned' magnetic remanence lies largely in a high coercivity, high Curie temperature magnetic fraction. Coercivities in excess of 1800 Oe (Figs. 3 and 5) and Curie temperatures between 660 and 700 °C (Figs. 4 and 6) point to hematite as the significant magnetic mineral, and this is confirmed by IRM build-up curves (Fig. 11). Slight drops of about 10% in $J/J_o$ between 550 and 600 °C indicate that small amounts of magnetite may also be present.

The means of the 20 reversed and 19 normal sites (entries 4 and 5, Table 2) are within 10° of being exactly reversed, and this provides evidence that later secondary magnetic components superimposed on an underlying magnetization are unimportant. However, this underlying magnetization is probably a chemical remanent magnetization, and it is therefore important to enquire when it was acquired. The well known fold test (Graham 1949) can discriminate between pre- and post-folding magnetization episodes, but in the present case there is not enough variation in the attitudes of the strata sampled to cause a significant difference in statistics when mean directions are calculated in the present and ancient coordinate frames (entries 1 and 2, Table 2).

Directional and intensity data for the main section strongly suggest that the strata were magnetized sequentially, either progressively upwards or progressively downwards. This is most readily achieved during diagenesis, either upon deposition or as lithification proceeds vertically through the sediment pile. Indeed, in a recent review, Baag and Helsley (1974, p. 3308) concluded that "most paleomagnetic literature concerning red beds supports the view that the stable primary NRM of red beds was acquired at and/or soon after deposition". Furthermore, these authors point out that the presence of a gradual trend of intermediate directions between opposed polarity zones favours syndepositional acquisition of remanence. Despite these favourable points, it is, of course, possible to devise later remagnetization schemes (*e.g.*, movement of an isotherm or chemical front) that could give rise to the observed facts, and the problem cannot be completely resolved by reference to the Stark Formation data alone. The possibly primary nature of the remanence was discussed further in a companion paper (Evans and Bingham 1975), where results from stratigraphically adjacent formations were described. There it is shown that the remanence does indeed pre-date the folding which, in turn, predates deposition of the overlying Et-then Group.

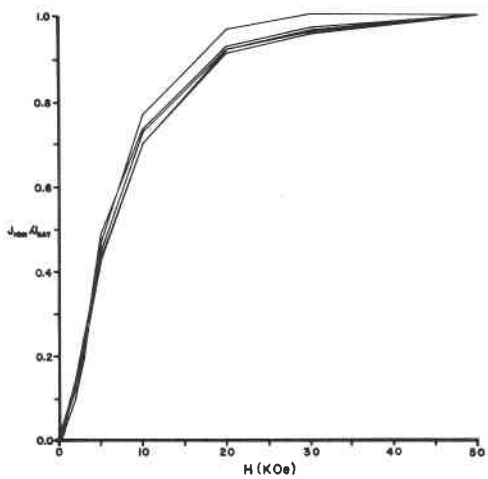



FIG. 11. Intensity of IRM normalized to the value achieved at 50 kOe. Specimens from sites SBA, SBF, SBL, SBN, and SBP of the main section.

*Pole positions*

The main section mean direction, which best represents the true mean paleomagnetic field direction during Stark times (entry 1, Table 2), yields a pole position at 145° W, 15° S in the equatorial mid-Pacific. Sites SDA, SDB, SEB, and SEC are not included, since they are closely associated with periods of geomagnetic instability, and they represent insufficient data to warrant the assumption that stable dipole positions had been achieved. A paleolatitude of 8° is implied and this is consistent with the abundant occurrence of halite casts. The pole position falls more than 60° to the west of both the Et-then Group pole (Irving *et al.* 1972) and the Kahochella Group pole (McMurry *et al.* 1973), which are known to lie stratigraphically above and below the Stark Formation, respectively. Taken at face value, the Stark result requires an additional large loop in the polar wander path for North America. Although the time period involved for the new loop (~ 200 m.y.) is probably long enough to accommodate it at recognized rates of polar wandering (*e.g.*

Creer *et al.* 1957), this added complexity in the total curve begins to make the polar wander curve more difficult to use as a stratigraphic or geotectonic tool. Therefore, it is worthwhile to consider possible alternatives.

(a) The Stark pole is not significantly different from the 'Elsonian pole' of 1300–1475 m.y. (Spall 1971; Evans *et al.* 1975), but an 'Elsonian' age is at variance with all the geological facts (see, *e.g.*, Hoffman 1973).

(b) It is always possible to explain any apparent anomaly in a polar wandering curve by supposing that for the time in question the geomagnetic field was non-dipolar in character. Evidence for the dipolar nature of the geomagnetic field during Stark or earlier times is lacking, but evidence exists for such behaviour at later times, the earliest of which is the Elsonian Pole (Evans *et al.* 1975). At the present time it is unwise to appeal to such *ad hoc* hypotheses until all simpler suggestions are firmly ruled out.

(c) Introducing an additional loop can be avoided by drawing the polar wander path through the Stark Formation anti-pole. Instead of producing a new loop across the Pacific, this alternative stretches the presently accepted northern South American loop 90° further eastward to a point near Khartoum, Sudan. In like manner, one could use the other end of the Ethen pole as the normal pole, thereby producing a westward loop over New Guinea. Either (or both) of these possibilities are viable suggestions, and a final decision will only be possible as more data become available.

(d) Perhaps some additional tectonic correction to the paleomagnetic data is necessary, such as rotation of a small block about a vertical axis. This mechanism has previously been proposed to account for divergent paleomagnetic poles (*e.g.* Fahrig and Larochelle 1972; Roy and Fahrig 1973). Such suggestions require extreme caution since, used arbitrarily, they can readily give rise to misleading interpretations. However, in the present case there exists evidence that such a rotation may have occurred. Examination of the geological map (Geological Survey of Canada, Map 1122A, 1968) indicates that the general trend of the East Arm Fold Belt is approximately NE–SW, but that in the area covered by Fig. 1 fold axes trend more E–W or even south of east. Quantitative analysis of bedding attitudes (Fig. 9) confirm this discrepancy. If this change in strike represents a tectonic bending by some 50–60°, then appropriate allowance must be made for it. The intriguing situation then arises that the paleomagnetic pole is shifted to a position much more consistent with those derived from North American rocks of similar age. For example, rotation by 60° about a vertical axis yields a pole at 83° W 18° S. A firm decision cannot be made on this problem until the possibility of rotation is tested by adequate sampling in areas where the tectonic 'grain' is conformable with that of the entire basin.

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BAAG, C. G. and HELSLEY, C. E. 1974. Evidence for penecontemporaneous magnetization of the Moenkopi Formation. J. Geophys. Res. **79**, pp. 3308–3320.

BARBETTI, M. and MCELHINNY, M. 1972. Evidence of a geomagnetic excursion 30,000 yr BP. Nature, **239**, pp. 327–330.

BARNES, F. Q. 1951. Snowdrift map-area, Northwest Territories. Geol. Surv. Can. Pap. 51–6, 30 p.

CREER, K. M., IRVING, E., and RUNCORN, S. K. 1957. Geophysical interpretation of paleomagnetic directions from Great Britain. Phil. Trans. R. Soc. London A, **250**, pp. 144–156.

CREER, K. M. and ISPIR. Y. 1970*a*. Palaeomagnetic and rock magnetic studies on Cenozoic basalts from Kyushu, Japan. Geophys. J. R. Astron. Soc. **20**, pp. 127–148.

——— 1970*b*. An interpretation of the behaviour of the geomagnetic field during polarity transitions. Phys. Earth Planet. Interiors, **2**, pp. 283–293.

CRUDEN, D. M. 1968. Methods of calculating the axes of cylindrical folds: a review. Geol. Soc. Am. Bull. **79**, pp. 143–148.

DOELL, R. R. and COX, A. 1972. The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle. *In*: Nature of the solid Earth. (E. C. Robertson, *Ed.*) McGraw-Hill, New York, pp. 245–284.

DUNN, J. R., FULLER, M., ITO, H., and SCHMIDT, V. A. 1971. Paleomagnetic study of a reversal of the earth's magnetic field. Science, **172**, pp. 840–845.

EVANS, M. E. and BINGHAM, D. K. 1976. Paleomagnetism of the Great Slave Supergroup, N.W.T., Canada: the Tochatwi Formation. Can. J. Earth Sci. (this issue).

EVANS, M. E., BINGHAM, D. K., and MCMURRY, E. W. 1975. New paleomagnetic results from the Upper Belt – Purcell Supergroup of Alberta. Can. J. Earth Sci. **12**, pp. 52–61.

FAHRIG, W. F. and LAROCHELLE, A. 1972. Paleomagnetism of the Michael Gabbro and possible evidence of the rotation of Makkovik Subprovince. Can. J. Earth Sci. **9**, pp. 1287–1296.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. R. Soc. London A, **217**, pp. 295–305.

GRAHAM, J. W. 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res. **54**, pp. 131–167.

HOFFMAN, P. F. 1968. Stratigraphy of the Lower Proterozoic (Aphebian), Great Slave Supergroup, East Arm of Great Slave Lake, District of Mackenzie. Geol. Surv. Can. Pap. 68-42, 93 p.

——— 1969. Proterozoic paleocurrents and depositional history of the East Arm fold belt, Great Slave Lake, Northwest Territories. Can. J. Earth Sci. **6**, pp. 441–462.

——— 1973. Evolution of an early Proterozoic continental margin: the Coronation geosyncline and associated autocogens of the northwestern Canadian shield. Phil. Trans. R. Soc. London A, **273**, pp. 547–581.

HOFFMAN, P. F., FRASER, J. A., and MCGLYNN, J. C. 1970. The Coronation geosyncline of Aphebian age, District of Mackenzie. Geol. Surv. Can. Pap. 70-40, pp. 201–212.

IRVING, E., MOLYNEUX, L., and RUNCORN, S. K. 1966. The analysis of remanent intensities and susceptibilities of rocks. Geophys. J. R. Astron. Soc. **10**, pp. 451–464.

IRVING, E., PARK, J. K., and MCGLYNN, J. C. 1972. Paleomagnetism of the Et-then Group and Mackenzie Diabase in the Great Slave Lake area. Can. J. Earth Sci. **9**, pp. 744–755.

KUKAL, Z. 1971. Geology of Recent sediments. Academic Press, Prague. 490 p.

MCELHINNY, M. W. 1964. Statistical significance of the fold test in palaeomagnetism. Geophys. J. R. Astron. Soc. **8**, pp. 338–340.

——— 1966. An improved method for demagnetizing rocks in alternating magnetic fields. Geophys. J. R. Astron. Soc. **10**, pp. 369–374.

——— 1970. Palaeomagnetism of the Cambrian purple sandstone from the Salt range, West Pakistan. Earth Planet. Sci. Lett. **8**, pp. 149–156.

——— 1971. Geomagnetic polarity transitions. Comments on Earth Sci.: Geophys. **1**, pp. 150–158.

MCMURRY, E. W., REID, A. B., and EVANS, M. E. 1973. A paleomagnetic study of the Kahochella Group, N.W.T., Canada. EOS Trans. A.G.U. **54**, 248 p.

MURTHY, G. S. 1969. Paleomagnetic studies in the Canadian shield. Unpubl. PhD thesis, Univ. of Alberta, Calgary, Alberta.

PATTON, B. J. and FITCH, J. L. 1962. Design of a room-size magnetic shield. J. Geophys. Res. **67**, pp. 1117–1121.

REID, A. B., 1972. A palaeomagnetic study at 1800 million years in Canada. Unpubl. PhD thesis, Univ. of Alberta, Calgary, Alberta.

ROY, J. L. and FAHRIG, W. F. 1973. The paleomagnetism of Seal and Croteau rocks from the Grenville Front, Labrador: polar wandering and tectonic implications. Can. J. Earth Sci. **10**, pp. 1279–1301.

SPALL, H. 1971. Precambrian apparent polar wandering: evidence from North America. Earth Planet. Sci. Lett. **10**, pp. 273–280.

STOCKWELL, C. H. 1936. Great Slave Lake, District of Mackenzie, Northwest Territories. Geol. Surv. Can. Map 378A.

TARLING, D. H. 1966. The magnetic intensity and susceptibility distribution in some Cenozoic and Jurassic basalts, Geophys. J. R. Astron. Soc. **11**, pp. 423–432.

VINCENZ, S. A. and BRUCKSHAW, L.McG. 1960. Note on the probability distribution of a small number of vectors. Proc. Camb. Phil. Soc. **56**, pp. 21–26.

WATKINS, N. D. and NOUGIER, J. 1973. Excursions and secular variation of the Brunhes epoch geomagnetic field in the Indian Ocean region. J. Geophys. Res. **78**, pp. 6060–6068.

WATSON, G. S. 1956*a*. A test for randomness of directions. Mon. Not. R. Astron. Soc., Geophys. Suppl. **7**, pp. 160–161.

——— 1956*b*. Analysis of dispersion on a sphere. Mon. Not. R. Astron. Soc., Geophys. Suppl. **7**, pp. 153–159.

——— 1960. More significance tests on the sphere. Biometrika, **47**, pp. 87–91.